

L11c: Markov Decision Processes

Value iteration and random rollout

CHEME 5800 Cornell University

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Objectives for Today

How should an agent choose its actions when each action changes both the reward it collects and the state it moves to?

Today's concepts

- Formulate a Markov decision process: define its states, actions, rewards, transition model, and discount factor, and the policy that chooses an action in each state.
- Solve for the optimal policy with value iteration: update the value of each state from the states it can reach next until the values stop changing.
- Estimate values with random rollouts: average the discounted rewards of simulated random trajectories, without writing down the transition probabilities.

Lecture: Markov Decision Processes (MDPs)

Lecture and Example

Lecture: Markov Decision Processes (MDPs)
In this lecture, we formulate Markov decision processes, solve them with value iteration, and estimate values with random rollouts.

Example: Random Rollout in the Lava Grid World
In this example, we estimate the value of every square in a lava grid world by averaging random rollouts. We then read a policy from the estimates, draw it, and follow it from a starting square to the charging station.

From Markov Chains to Decisions

A Markov decision process (MDP) extends the Markov chain by adding actions and rewards.

- A Markov chain evolves passively, according to fixed transition probabilities.
- In an MDP, an agent chooses actions that influence the state transitions and generate rewards.

MDPs are the foundation for sequential decision-making under uncertainty: the agent must make good decisions, one after another, in a stochastic environment.

MDP Components

An MDP is defined by the tuple $(\mathcal{S}, \mathcal{A}, R(s,a), T(s' \mid s,a), \gamma)$:

- State space $\mathcal{S}$: all possible states s of the system.
- Action space $\mathcal{A}$: all possible actions a; the actions available in state s are $\mathcal{A}_s \subseteq \mathcal{A}$.
- Reward function $R(s,a)$: the expected immediate reward of action a in state s. When the outcome is random, $R(s,a)$ averages over the outcomes; r is the reward actually received on a single step.
- Transition model $T(s' \mid s,a) = P(s_{t+1} = s' \mid s_t = s, a_t = a)$: the probability that action a in state s leads to state s'.
- Discount factor $\gamma \in [0,1)$: weights future rewards against immediate ones; $\gamma = 0$ cares only about now, and $\gamma \to 1$ values long-term returns.

The Agent's Goal: A Policy

The agent's goal is a policy $\pi : \mathcal{S} \to \mathcal{A}$, with $\pi(s) = a$, that maps states to actions while maximizing the expected cumulative discounted reward:

$$\mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t)\right].$$

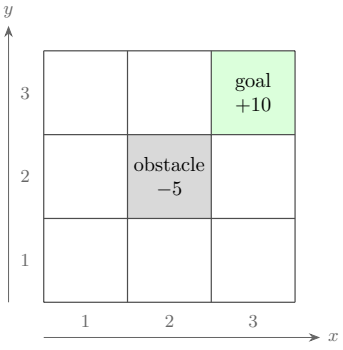
Previously, π denoted a state distribution; here it is a policy.

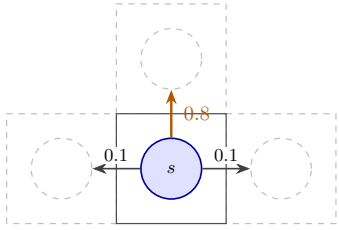
Example: Grid World Navigation

A robot navigates a 3×3 grid to reach a goal while avoiding an obstacle and minimizing travel time:

- States: the eight squares (x, y), $x, y \in \{1, 2, 3\}$, other than the obstacle at $(2, 2)$; x increases to the right and y upward.
- Actions: up, down, left, right; moves off the grid are unavailable, so $\mathcal{A}_s \subsetneq \mathcal{A}$ on the edges.
- Rewards: $+10$ for entering the goal at $(3, 3)$, -5 for a bump into the obstacle, and -1 for every other outcome, including a slip off the grid.
- Transitions: slippery terrain; the intended move succeeds with probability 0.8, and each perpendicular move happens with probability 0.1.
- Discount factor $\gamma = 0.9$: the $+10$ goal outweighs the -1 step costs, so the robot pursues the goal.

Grid World: The Layout





action $a =$ up; a move off the grid or into the obstacle leaves the robot in place

Because moves are random, $R(s, a)$ averages over where the robot actually moves. A bump or a slip off the grid leaves it in place, and the goal is terminal, so it earns zero further reward after entry.

Grid Worlds in Chemical Engineering

Divide a process's operating window into a grid, with two operating variables, such as temperature and feed flow rate, on the axes:

- States: operating points; we want to move the process to a point that meets a new product specification.
- Actions: raise or lower one operating variable by one step.
- Transitions: each action moves to a neighboring point; disturbances play the role of slips.
- Rewards: a large reward for meeting the specification, large negative rewards for violating a constraint, and a small cost per step of off-spec product.

The optimal policy says which variable to change at each operating point.

Value Iteration

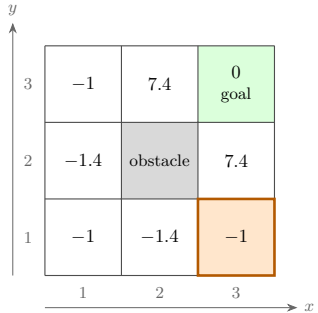
Value iteration is a dynamic programming algorithm: it builds the value of each state from the values of the states that can follow it. It repeats the Bellman backup:

$$U_{k+1}(s) = \max_{a \in \mathcal{A}_s} \left(R(s,a) + \gamma \sum_{s' \in \mathcal{S}} T(s' \mid s, a)\, U_k(s') \right)$$

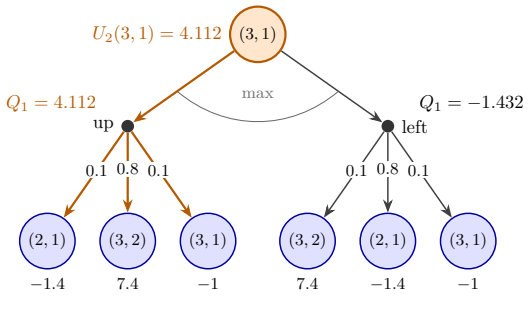
- For $0 \leq \gamma < 1$, $U_k(s) \to U^*(s)$ as $k \to \infty$.
- The optimal value function $U^*(s)$ is the maximum expected cumulative discounted reward achievable from state s under the best possible policy.

Worked Example: One Bellman Backup

Update the corner $s = (3, 1)$: only up and left keep the robot on the grid, and every outcome of either action earns -1. Starting from $U_0 = 0$, one sweep leaves these values:



values $U_1(s)$ after one sweep



numbers below the states are $U_1(s')$; every move earns $r = -1$; $\gamma = 0.9$

Where Do 7.4 and 0 Come From?

The $+10$ is earned on the move into the goal, so it counts toward the square the robot moves from. From $(3, 2)$, moving up gives

$$0.8\,(+10) + 0.1\,(-5) + 0.1\,(-1) = 7.4,$$

reaching the goal with probability 0.8, bumping the obstacle with probability 0.1, and slipping off the grid with probability 0.1.

The goal itself is terminal: nothing is left to earn there, so its value stays 0.

Worked Example: Comparing the Two Actions

$Q_1(s, a)$ is the expected return of taking a and then using the values U_1. Up reaches $(3, 2)$ with probability 0.8, slips to $(2, 1)$ or stays with probability 0.1 each; left swaps the two neighbors:

$$Q_1((3,1), \text{up}) = -1 + 0.9(0.8(7.4) + 0.1(-1.4) + 0.1(-1)) = 4.112,$$
$$Q_1((3,1), \text{left}) = -1 + 0.9(0.8(-1.4) + 0.1(7.4) + 0.1(-1)) = -1.432.$$

The Bellman backup takes the larger:

$$\boxed{U_2(3,1) = \max\{4.112,\, -1.432\} = 4.112}$$

The greedy action is up. Every state in the next sweep uses the previous sweep's values in the same way.

Value Iteration Algorithm

Initialize: Given an MDP, a tolerance $\epsilon > 0$, and a maximum number of iterations $k_{\max}$, set $k \leftarrow 0$, $U_0(s) \leftarrow 0$ for all $s \in \mathcal{S}$, and `converged` $\leftarrow$ `false`.

While not `converged` **do**:

1. For each state $s \in \mathcal{S}$, compute the Bellman backup $U_{k+1}(s)$.
2. Check whether to stop:
   - If $\max_{s \in \mathcal{S}} |U_{k+1}(s) - U_k(s)| \leq \epsilon$ **then** set `converged` $\leftarrow$ `true` and return $U^* \leftarrow U_{k+1}$.
   - Otherwise, if $k + 1 \geq k_{\max}$ **then** return U_{k+1} with `converged` $=$ `false` and a warning.
3. Increment the iteration counter, $k \leftarrow k + 1$.

Extract Policy: for each state $s \in \mathcal{S}$,

$$\pi^*(s) \leftarrow \arg\max_{a \in \mathcal{A}_s} \left(R(s,a) + \gamma \sum_{s' \in \mathcal{S}} T(s' \mid s, a)\, U^*(s') \right).$$

Value Iteration: Cost

Each iteration costs $\mathcal{O}\left(|\mathcal{S}|^2\,|\mathcal{A}|\right)$ operations:

- for each of the $|\mathcal{S}|$ states, we evaluate $|\mathcal{A}|$ actions,
- and each action sums over $|\mathcal{S}|$ possible next states.

The number of iterations depends on γ and ϵ: each iteration shrinks the largest error in the value function by at least a factor of γ.

Value iteration is tractable for thousands of states, but challenging for continuous or very large discrete state spaces. There, we need function approximation or sampling-based methods.

Random Rollout

Random rollout estimates $U(s)$ by simulation:

- Start at s, choose actions at random, and add up the discounted rewards.
- Average over many rollouts to get $\hat{U}(s)$.

Nothing in that loop reads $T(s' \mid s, a)$ or $R(s, a)$. The environment only has to report a next state and a reward, so rollout runs where value iteration cannot.

Random Rollout Algorithm

Initialize: Given a simulated environment, $\gamma \in [0, 1)$, a maximum depth $d \geq 1$, and a number of rollouts $N \geq 1$, set $\hat{U}(s) \leftarrow 0$ and $n(s) \leftarrow 0$ for all $s \in \mathcal{S}$.

For $i = 1$ to N **do**:

1. Choose a starting state s_0; set $s \leftarrow s_0$, $t \leftarrow 0$, $G \leftarrow 0$, and `terminated` $\leftarrow$ `true` if s_0 is terminal, otherwise `false`.
2. While not `terminated` **do**:
   - Select a uniformly at random from $\mathcal{A}_s$; execute it and observe r and s'.
   - Update $G \leftarrow G + \gamma^t r$ and $t \leftarrow t + 1$.
   - If $t \geq d$ or s' is terminal **then** set `terminated` $\leftarrow$ `true`; otherwise, $s \leftarrow s'$.
3. Update the estimate with an incremental mean:

$$n(s_0) \leftarrow n(s_0) + 1, \qquad \hat{U}(s_0) \leftarrow \hat{U}(s_0) + \frac{1}{n(s_0)} \left(G - \hat{U}(s_0) \right).$$

Output: $\hat{U}(s)$ for every state that served as a starting state.

Random Rollout: Cost and Accuracy

Random rollout costs $\mathcal{O}(N \cdot d)$ simulated steps.

- Apart from one estimate per state, that cost does not grow with $|\mathcal{S}|$, which makes rollout usable on large or continuous state spaces.
- Accuracy comes slowly: the standard error of a sample mean falls as $\mathcal{O}(1/\sqrt{n(s)})$, so ten times the rollouts buys about three times the precision.

Example: Random Rollout in the Lava Grid World

In this example, a robot must reach its charging station without falling into a lava pit:

1. Build the lava grid world as an MDP, and shape the rewards so moves toward the station pay a little more.
2. Estimate each square's value by averaging random rollouts.
3. Read a policy from the estimates with $Q(s, a)$, and follow it from a start square.

The policy improves on random moves, but it is not the optimal policy that value iteration computes.

Example: Random rollout in the lava grid world

Lab: Value Iteration in a Grid World

In the lab, a robot must reach a charging station while avoiding lava pits:

- we build the MDP, its reward matrix and transition array, for the robot's grid world,
- we write our own value iteration with a convergence check and compare it with the course package's version, which runs a fixed number of sweeps,
- we extract the optimal policy, follow the robot's path, and see how the discount factor changes its behavior.

Lab: Value iteration in a grid world

Summary

We added actions and rewards to Markov chains, which lets an agent make optimal sequential decisions.

Key takeaways

- **MDPs.** Actions shape where the process goes, rewards score each choice, and a policy assigns an action to every state.
- **Value iteration.** Repeated Bellman backups give the optimal values; the action that achieves each maximum is the optimal policy.
- **Random rollout.** Averaging simulated random trajectories estimates how well random actions do, without the transition probabilities.

Value iteration needs the transition model; random rollout needs only a way to simulate the environment.